CoRe-TFM: Selective Four-View Probability Reconciliation for Tabular Foundation Models

Badampudi Agasthya Anirudh[1*]

[1*]Integrated M.Tech Data Science, [Institution to be completed], India.

Corresponding author(s). E-mail(s): anirudhbadampudi@gmail.com;

Abstract

Tabular foundation models (TFMs) can expose direct marginal and conditional probabilities that do not correspond to a common joint distribution. We study post-hoc reconciliation of four views without retraining the underlying model. Hard CoRe exactly preserves direct marginals, Soft CoRe relaxes marginal preservation through KL penalties, and Selective CoRe uses held-out labels to choose among raw factorizations, pooling, and reconciliation. Known-truth experiments show that hard preservation helps when direct marginals are reliable and hurts when they are not. Across 100 heterogeneous controlled tasks, Selective CoRe beats arithmetic pooling by **0.00666** mean NLL. The conclusion reverses in a bounded real-model benchmark covering ten data sets, five folds, and two TFMs: Selective CoRe is worse than arithmetic by **0.001831** NLL (95% bootstrap CI $[\mathbf{0.000548, 0.003301}]$; Wilcoxon $\boldsymbol{p = 0.0273}$), while reducing marginal distortion. Effects reverse by upstream model. Probability reconciliation is therefore a reliability-dependent consistency–accuracy trade-off, and finite-validation selection is a principal deployment limitation.

Keywords: tabular foundation models, probabilistic consistency, probability reconciliation, calibration, proper scoring rules, Sinkhorn scaling

1 Introduction

Tabular foundation models (TFMs) have rapidly become competitive general-purpose predictors for structured data. Models such as TabPFN-3 (Grinsztajn et al. 2026) and TabICLv2 (Qu et al. 2026) use large-scale pretraining and in-context inference to produce task-specific predictions without conventional task-by-task optimization. Their probability outputs are attractive beyond ordinary classification accuracy: they can be used for uncertainty-sensitive decisions, conditional queries, sequential simulation, imputation, and multivariate scenario generation.

These uses expose a structural question that is largely invisible in standard single-target benchmarks. Suppose a frozen predictor is queried for two categorical targets A and B. The same model can produce direct marginals $p(A \mid X)$ and $p(B \mid X)$, together with conditional predictions $p(A \mid B, X)$ and $p(B \mid A, X)$. If all four predictions were views of one coherent joint distribution, two elementary requirements would hold: marginalizing an appropriate conditional construction would recover the corresponding direct marginal, and the two chain-rule factorizations would define the same joint.

Kloetergens et al. (Kloetergens et al. 2026) recently tested these requirements and found systematic marginalization and factorization violations across evaluated TFMs and datasets, for both classification and regression. Their study establishes that TFM outputs cannot in general be interpreted as arbitrary views of a single joint. It also leaves open a practically important question: can incompatible predictions be reconciled after inference without retraining, and can this be done without discarding predictive information?

That second question is harder than it initially appears. Constructing any single normalized joint makes its own marginals and conditionals mutually compatible by definition. Consequently, "zero inconsistency after repair" is not evidence that a repair is useful. A coherent distribution may be worse calibrated, have worse proper scores, or erase dependence. Recent TFM work reinforces this distinction. Proper-scoring benchmarks can change model rankings (Landsgesell and Knoll 2026b,a); uncertainty benchmarking finds strong point prediction alongside weaker reliability properties (Costa et al. 2026); and distribution shift remains a separate challenge (Loza et al. 2026). Post-hoc probability adjustment has also already been studied for label shift through DistPFN (Lee et al. 2026). CoRe-TFM therefore targets neither generic calibration nor generic posterior adjustment. It targets a specific compatibility problem among multiple predictions produced by the same TFM.

We formulate four-view reconciliation as a family of information projections. Two autoregressive factorizations are first treated as competing joint views. Their arithmetic and geometric pools have KL-barycenter interpretations. Hard CoRe takes a consensus and projects it onto the transportation polytope defined by the two direct marginals. Soft CoRe replaces the hard marginal constraints with KL penalties, yielding a generalized matrix-scaling problem. Selective CoRe treats reconciliation as a decision: a validation set may choose an original factorization, a pool, or a repair rather than forcing coherence on every problem.

The empirical design is diagnostic rather than winner-takes-all. A surrogate data-generating process and exact-view perturbations provide known truth; a heterogeneous 100-task mixture tests adaptive selection; and a completed bounded-context matrix evaluates two released TFMs across ten data sets. Together these experiments identify both the regimes in which reconciliation helps and the finite-validation regimes in which simple pooling is stronger.

Our contributions are:

1. a four-view formulation of TFM probability reconciliation that distinguishes compatibility from calibration and predictive correctness;

2. Hard and Soft CoRe objectives, including generalized Sinkhorn-style updates for the soft marginal-penalty problem;
3. a Selective CoRe protocol whose finite-validation selection cost is measured explicitly;
4. exact-view and heterogeneous-task benchmarks that identify when marginal preservation or adaptive selection helps;
5. a completed bounded-context benchmark over ten data sets, two released TFMs, and a non-TFM boundary baseline; and
6. an auditable evaluation framework with fold outputs, checksums, ablations, sensitivity analyses, and dataset-blocked inference.

2 Related work

2.1 Tabular foundation models

The current TFM landscape evolves quickly. TabPFN-3 scales prior-data fitted prediction to substantially larger tables and introduces test-time compute scaling (Grinsztajn et al. 2026). TabICLv2 reports scalable open inference code and released weights for classification and regression (Qu et al. 2026). TabFM is another recent zero-shot tabular foundation model release (Google Research 2026). These models differ in pretraining distributions, architectures, scaling strategies, and inference constraints, making cross-model consistency evaluation particularly relevant.

2.2 Probabilistic reliability of TFMs

The most direct precursor is Kloetergens et al. (2026), who define marginalization and factorization consistency tests and report violations for all evaluated model–dataset combinations. CoRe-TFM addresses the post-hoc reconciliation direction that follows from this diagnosis. Related reliability studies ask different questions. Costa et al. (2026) compare predictive performance with uncertainty quality across 112 datasets. Landsgesell and Knoll (2026b) and Landsgesell and Knoll (2026a) argue for proper scoring rules when TFMs produce predictive distributions. Loza et al. (2026) evaluate TFM degradation under distribution shift. DistPFN (Lee et al. 2026) adjusts TFM class probabilities at test time to mitigate label shift. These results motivate evaluating reconciliation with proper scores and calibration, but none makes probability coherence a surrogate for those properties.

2.3 Compatibility of conditional distributions

Compatibility of separately specified conditionals predates foundation models. Incompatible conditional distributions can still define useful compromises; for example, Mure (2017) studies stationary compromise distributions induced by pseudo-Gibbs updates. A related modern setting appears in conditional neural processes: Young (2026) quantifies a conditioning-consistency gap rather than assuming consistency. We therefore do not claim to introduce the general notion of conditional compatibility. The contribution is to instantiate and evaluate reconciliation for the four predictive views naturally exposed by TFMs.

2.4 Information projection and matrix scaling

Hard CoRe is an information projection onto a convex set with prescribed marginals. Iterative proportional fitting and Bregman projections are classical tools for such problems (Benamou et al. 2015; Cuturi 2013). Relaxing marginal equality with KL penalties leads to generalized scaling forms closely related to unbalanced transport algorithms (Chizat et al. 2018). The optimization machinery is prior art. We use it because it provides a transparent and efficient solver for the specific TFM reconciliation objective.

3 Problem formulation

Let $X = x$ denote test covariates and let $A \in \{1, \ldots, K_A\}$ and $B \in \{1, \ldots, K_B\}$ be categorical targets. For a fixed predictor, suppressing its context dataset in the notation, obtain four probability views:

$$p_A(a) = \hat{p}(A = a \mid X = x), \qquad p_B(b) = \hat{p}(B = b \mid X = x), \tag{1}$$
$$p_{A|B}(a \mid b) = \hat{p}(A = a \mid B = b, X = x), \quad p_{B|A}(b \mid a) = \hat{p}(B = b \mid A = a, X = x). \tag{2}$$

They imply two chain-rule joints:

$$J^{B\to A}(a, b) = p_B(b)p_{A|B}(a \mid b), \tag{3}$$
$$J^{A\to B}(a, b) = p_A(a)p_{B|A}(b \mid a). \tag{4}$$

Under exact compatibility, $J^{B\to A} = J^{A\to B}$. We quantify factorization inconsistency at a test point by

$$\Delta_{\text{fac}}(x) = \text{TV}(J^{B\to A}, J^{A\to B}) = \frac{1}{2}\sum_{a,b} |J^{B\to A}(a, b) - J^{A\to B}(a, b)|. \tag{5}$$

Each chain construction preserves one direct marginal automatically but may violate the other. We also measure

$$\Delta_A(x) = \text{TV}\left(\sum_b J^{B\to A}(a, b), p_A(a)\right), \tag{6}$$
$$\Delta_B(x) = \text{TV}\left(\sum_a J^{A\to B}(a, b), p_B(b)\right). \tag{7}$$

The goal is not simply to make these quantities zero. We seek a joint Q that is internally compatible while controlling predictive degradation relative to the original views.

4 CoRe-TFM

4.1 Consensus joints as KL barycenters

For weight $w \in [0, 1]$, the arithmetic pool $Q_{\text{arith}} = wJ^{B \to A} + (1-w)J^{A \to B}$ minimizes a weighted sum of forward KL divergences from the input distributions to Q. Conversely, the normalized geometric pool

$$M_w(a, b) = \frac{J^{B \to A}(a, b)^w J^{A \to B}(a, b)^{1-w}}{\sum_{a',b'} J^{B \to A}(a', b')^w J^{A \to B}(a', b')^{1-w}} \tag{8}$$

minimizes the corresponding weighted reverse-KL objective in Q. Both are baselines; the geometric pool is also the reference measure for CoRe.

4.2 Hard marginal-preserving reconciliation

Let

$$\mathcal{T}(p_A, p_B) = \{Q \geq 0 : \sum_b Q(a, b) = p_A(a), \ \sum_a Q(a, b) = p_B(b)\} \tag{9}$$

be the transportation polytope determined by the direct predictions. Hard CoRe solves

$$Q_{\text{hard}} = \arg \min_{Q \in \mathcal{T}(p_A, p_B)} \text{KL}(Q \| M_w). \tag{10}$$

Proposition 1 (Feasibility and marginal preservation)**.** *For any probability vectors p_A and p_B, the feasible set in Eq. (10) is non-empty. Every solution preserves the two direct marginal predictions exactly.*

Proof The independent joint $Q_0(a, b) = p_A(a)p_B(b)$ belongs to $\mathcal{T}(p_A, p_B)$. The second statement follows from the constraints. □

Proposition 2 (Uniqueness under positive support)**.** *If $M_w(a, b) > 0$ for all states and the feasible set contains a strictly positive point, Eq. (10) has a unique minimizer.*

Proof The transportation polytope is convex and compact. $\text{KL}(Q \| M_w)$ is strictly convex in Q over positive support, so a minimizer exists and is unique. □

The solution can be computed by alternating row and column scaling, equivalently iterative proportional fitting/Bregman projection for this finite problem (Benamou et al. 2015).

4.3 Soft four-view reconciliation

Hard preservation encodes a strong assumption: direct marginals are more trustworthy than conditional views. We relax it through

$$Q_{\text{soft}} = \arg \min_Q w \, \text{KL}(Q \| J^{B \to A}) + (1 - w) \, \text{KL}(Q \| J^{A \to B})$$

$$+ \lambda_A \,\mathrm{KL}(Q_A \| p_A) + \lambda_B \,\mathrm{KL}(Q_B \| p_B). \tag{11}$$

Up to constants independent of Q, the first two terms equal $\mathrm{KL}(Q \| M_w)$. By the KL chain rule,

$$\mathrm{KL}(Q \| J^{B\to A}) = \mathrm{KL}(Q_B \| p_B) + \mathbb{E}_{b\sim Q_B} \,\mathrm{KL}(Q_{A|B=b} \| p_{A|B=b}), \tag{12}$$

$$\mathrm{KL}(Q \| J^{A\to B}) = \mathrm{KL}(Q_A \| p_A) + \mathbb{E}_{a\sim Q_A} \,\mathrm{KL}(Q_{B|A=a} \| p_{B|A=a}). \tag{13}$$

Thus Eq. (11) reconciles both direct marginals and both conditional families rather than merely averaging two matrices.

4.4 Generalized Sinkhorn-style updates

For positive M_w, the optimum has a scaling form $Q \propto \mathrm{diag}(u) M_w \mathrm{diag}(v)$. Alternating first-order updates are

$$u \leftarrow \left(\frac{p_A}{M_w v}\right)^{\tau_A}, \qquad \tau_A = \frac{\lambda_A}{1+\lambda_A}, \tag{14}$$

$$v \leftarrow \left(\frac{p_B}{M_w^\top u}\right)^{\tau_B}, \qquad \tau_B = \frac{\lambda_B}{1+\lambda_B}. \tag{15}$$

These are a specialization of generalized diagonal scaling used for KL-relaxed marginal problems (Chizat et al. 2018). An independent exponentiated-gradient implementation and a generic constrained optimizer on small instances are retained as numerical cross-checks.

4.5 Reliability weighting and Selective CoRe

Equal $w = 1/2$ assumes symmetric reliability of chain orders. We choose w from a small grid using validation joint log loss, and select soft marginal penalties on the same inner validation split. Selective CoRe extends this by choosing among an original factorization, arithmetic/geometric pooling, hard reconciliation, and soft reconciliation before outer-test evaluation.

Let L_{sel} be the test joint NLL of the validation-selected original factorization and L_Q that of candidate Q. Define

$$\mathrm{Tax}(Q) = L_Q - L_{\mathrm{sel}}, \qquad \mathrm{Dividend}(Q) = L_{\mathrm{sel}} - L_Q. \tag{16}$$

A positive dividend is evidence that reconciliation adds predictive value beyond selecting a factorization order. This prevents a purely algebraic gain in coherence from being reported as predictive improvement.

5 Experimental design

5.1 Evaluation principles

Proper scoring rules are central because the output is a probability distribution, not only a class decision (Gneiting and Raftery 2007). We report joint NLL and Brier score; marginal NLL/Brier/accuracy; joint and marginal expected calibration error (ECE); NLL and Brier scores for the two conditional distributions induced by each candidate joint; factorization and marginalization TV; reconciliation and marginal distortion; dependence fidelity through mutual-information error; and runtime. ECE is treated as descriptive because it depends on binning and is not a proper scoring rule (Guo et al. 2017).

For synthetic experiments with known truth $P^*(A, B \mid X)$, we additionally report TV and Jensen–Shannon divergence to truth and exact expected log loss

$$\mathcal{L}_{P^*}(Q) = -\mathbb{E}_X \sum_{a,b} P^*(a, b \mid X) \log Q(a, b \mid X). \tag{17}$$

Using the expected score rather than sampled test labels removes evaluation noise and makes method differences attributable to the predicted distributions themselves.

5.2 Surrogate DGP benchmark

Our first controlled benchmark uses a multiclass DGP with a known chain factorization. Features are Gaussian, $A \mid X$ follows a softmax model, and $B \mid A, X$ follows another softmax with dependence parameter γ. Smooth nonlinear feature terms create model misspecification for a multinomial logistic-regression surrogate. One factor is varied at a time: $\gamma \in \{0, 0.5, 1.5, 3\}$; $n \in \{250, 1000, 5000\}$; $d \in \{5, 20, 50\}$; $K_A = K_B \in \{2, 3, 5\}$; and three imbalance levels. Each condition uses 10 independent seeds and an inner validation split for all adaptive choices. The surrogate exists only to validate the reconciliation protocol without a foundation-model checkpoint.

5.3 Exact-view perturbation benchmark

The surrogate benchmark entangles estimation error across all four model queries. We therefore introduce a complementary mechanism-identification experiment. For each seed we generate an exact multiclass $P^*(A, B \mid X)$, derive its compatible p^*_A, p^*_B, $p^*_{A|B}$ and $p^*_{B|A}$, and independently perturb each view in log-probability space. After renormalization, this produces incompatible but controlled inputs to the reconciliation methods. The test score is the exact expectation under unperturbed P^*.

We study six regimes with 50 paired seeds: (i) uniformly low noise; (ii) reliable direct marginals and noisy conditionals; (iii) noisy marginals and reliable conditionals; (iv) a reliable $B \to A$ factorization; (v) a reliable $A \to B$ factorization; and (vi) uniformly high noise. This benchmark is intentionally not a model simulator. Its purpose is to test the assumptions encoded by the reconciliation objective.

Table 1 Classification target pairs used in the completed bounded benchmark.

Dataset	OpenML ID	Rows	Target A	Target B
Anneal	46906	898	classes	steel
Credit	46918	1,000	good/bad customer	checking status
Phishing	46963	1,353	WebsiteType	SFH
MIC	46980	1,674	LET_IS	ASP_S_n
Customer	46938	1,723	bad-client target	education
Car	40975	1,728	class	safety
Marketing	46940	2,216	Response	Marital_Status
Wine	40691+40498	6,497	quality class	color
Nursery	26	12,960	class	health
Diamonds	42225	53,940	cut	color

5.4 Statistical protocol

All adaptive decisions use validation data disjoint from test observations. In the DGP sweep, paired effects are computed within seed and summarized with bootstrap confidence intervals. In the exact-view study, four comparisons are pre-specified per regime: Hard CoRe versus arithmetic pooling, Adaptive Soft versus arithmetic, Adaptive Soft versus Hard CoRe, and Adaptive Soft versus validation-selected order. We use two-sided paired Wilcoxon signed-rank tests and Holm correction within each regime. Effect direction and magnitude are reported alongside corrected p-values rather than treating significance as the sole criterion.

5.5 Bounded real-model benchmark protocol

The completed validation layer uses TabICLv2 2.1.1 and TabPFN 8.4.0 as the primary TFMs, plus CatBoost 1.2.10 as a non-TFM boundary baseline. Each of five deterministic outer folds uses at most 256 training, 52 validation, and 128 test observations. TabFM was excluded before comparative outcome inspection because its released implementation was operationally incompatible with the bounded Colab protocol and prohibitively slow; CatBoost is not counted as its replacement in TFM claims. Primary inference averages the two TFMs within each dataset and uses the ten datasets as paired units.

6 Controlled results

6.1 Reconciliation value changes with dependence and sample size

The surrogate benchmark rejects a universal-ranking interpretation. At strong dependence ($\gamma = 3$), Adaptive Soft CoRe obtains mean joint NLL 1.5854, compared with 1.6082 for arithmetic pooling and 1.6048 for Hard CoRe. The validation-selected original order obtains 1.5865, showing that adaptive repair is competitive but that the

Table 2 Best mean exact expected NLL in each exact-view perturbation regime (50 paired seeds). Lower is better.

Perturbation regime	Best method	Expected NLL
Uniform low noise	Geometric pool	1.478604
Direct marginals reliable	Soft CoRe ($\lambda = 10$)	1.486760
Conditionals reliable	Arithmetic pool	1.508830
$B \to A$ factorization reliable	$J^{B\to A}$	1.475308
$A \to B$ factorization reliable	$J^{A\to B}$	1.475326
All four views noisy	Arithmetic pool	1.538679

available information can also be captured by choosing a good factorization. The independent joint is much worse (1.7435), despite being perfectly coherent and exactly preserving both direct marginals. Coherence alone is therefore insufficient.

At $n = 250$, arithmetic pooling is stronger than adaptive repair (mean NLL 2.0206 versus 2.0416 for Adaptive Soft), whereas at $n = 5000$ weighted geometric pooling, Selective CoRe, and Adaptive Soft are close. This suggests that validation-driven complexity can impose a selection cost in small samples.

A second diagnostic concerns whether inconsistency size predicts repair utility. Across the 40 seed–condition cells of the dependence sweep, factorization TV has essentially no rank association with the Adaptive-Soft dividend over a validation-selected original order ($\rho \approx 0.065$). It is more strongly associated with Adaptive Soft's advantage over unweighted arithmetic pooling ($\rho \approx 0.422$, $p \approx 0.0067$). Thus large inconsistency can indicate that symmetric averaging is inadequate, but it does not establish that repair is better than trusting the better factorization.

6.2 Exact-view perturbations identify the assumption behind hard preservation

Table 2 and Fig. 1 summarize the mechanism-identification study. The central result is not that one method dominates, but that method rankings follow the reliability structure of the four inputs.

When direct marginals are reliable and conditional views are noisy, Hard CoRe improves over arithmetic pooling by 0.01327 NLL on average (Hard minus arithmetic), with Holm-adjusted paired Wilcoxon $p = 7.11 \times 10^{-15}$. Soft CoRe with a strong marginal penalty ($\lambda = 10$) is marginally better still. This is the regime that most closely matches the modeling assumption behind Hard CoRe.

The conclusion reverses when conditionals are reliable but direct marginals are noisy. Hard CoRe is worse than arithmetic by 0.15048 NLL on average ($p_{\mathrm{Holm}} = 7.11 \times 10^{-15}$). Adaptive Soft is 0.14660 better than Hard CoRe but remains 0.00388 worse than arithmetic. Exact marginal preservation can therefore amplify a bad marginal forecast rather than repair the joint.

Directional regimes provide a third sanity check. When the $B \to A$ chain is reliable, $J^{B\to A}$ and the validation-selected order are best; Adaptive Soft is only 0.00054 NLL worse than the selected order. The analogous result holds for $J^{A\to B}$. A reconciliation

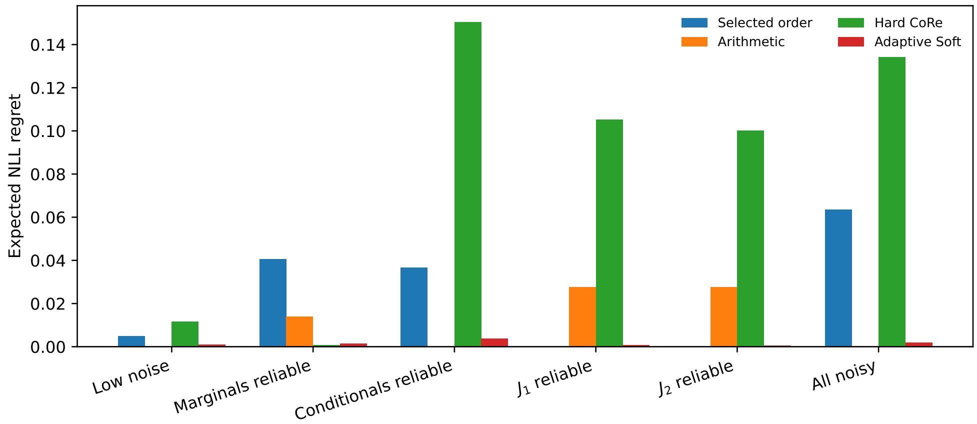

Fig. 1 Expected NLL regret relative to the best method within each exact-view perturbation regime. Strong marginal preservation is useful when direct marginals are reliable, but can incur a large cost when they are noisy.

algorithm cannot create information absent from its inputs, so a correct policy should be able to retain a trustworthy direction. This motivates Selective CoRe rather than mandatory repair.

Finally, under symmetric low or high noise, arithmetic/geometric pooling is difficult to beat. This negative result matters: averaging can be statistically efficient when input errors are roughly exchangeable, and a more structured projection is not justified solely because the inputs disagree.

6.3 Adaptive selection succeeds in heterogeneous controlled tasks

Across 100 tasks with independently varying view reliability, Selective CoRe obtains mean exact NLL 1.483684, compared with 1.490344 for the best fixed method, arithmetic pooling. The paired difference is -0.006660; Selective CoRe wins 68/100 tasks and the two-sided Wilcoxon p-value is 1.25×10^{-11}. Its mean regret to a non-deployable full-family oracle is 0.000710. It selects Soft CoRe 38 times, geometric pooling 33 times, a raw order 19 times, and arithmetic pooling 10 times.

A separate 50-task study increases nested validation size from 100 to 800 observations. Mean full-oracle regret decreases from 0.005602 to 0.001155, while mean gain over arithmetic increases from 0.000670 to 0.005118. This monotone controlled result does not imply that the same pattern must hold with the much smaller validation sets in the real benchmark.

6.4 Hard versus soft reconciliation is a reliability question

The exact-view study provides an empirical interpretation of the marginal penalty. Large λ expresses trust in direct marginals; $\lambda \to \infty$ approaches hard preservation, whereas small λ allows the reconciled joint to move its marginals toward conditional evidence. The best λ is therefore not merely a numerical tuning parameter. It estimates

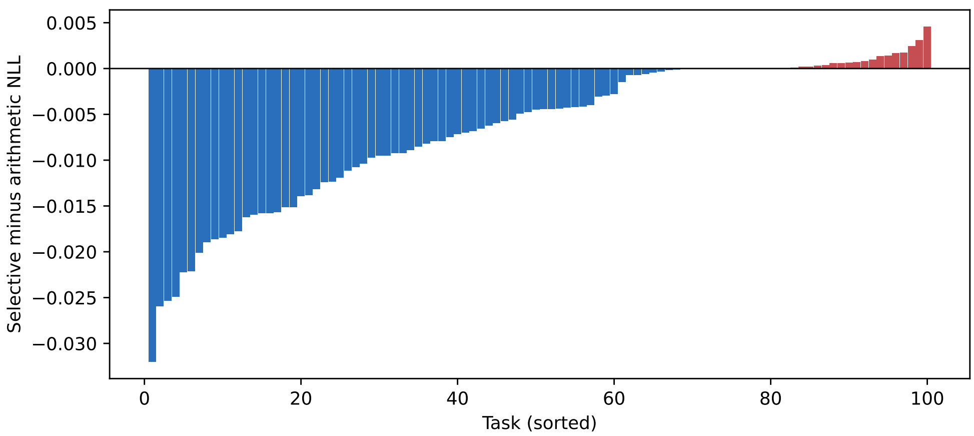

Fig. 2 Task-level Selective-CoRe effect relative to arithmetic pooling across 100 controlled reliability tasks. Negative values favor Selective CoRe.

a task-specific reliability trade-off among four TFM views. This interpretation explains why validation-adaptive penalties are preferable to a universal constant.

6.5 Directional marginal defects are not correctness signals

A natural label-free heuristic is to trust the factorization whose implied missing marginal is closer to its direct prediction. We tested this rule in the surrogate DGP. Its direction-selection accuracy weakens toward or below chance under stronger dependence. We therefore retain directional defect weighting only as a negative ablation. A small internal-consistency defect is not evidence that a direction is closer to P^*.

6.6 Computational overhead

On a local batch of 5,000 5×5 joints, the generalized scaling solver reached the same soft objective as an independently implemented exponentiated-gradient optimizer while requiring approximately 0.022 seconds versus 0.101 seconds. The intended interpretation is practical rather than algorithmic novelty: reconciliation is lightweight relative to foundation-model inference. Formal optimization novelty is not claimed because generalized Sinkhorn methods for KL-relaxed marginal problems are established (Chizat et al. 2018).

7 Bounded real-model results

The completed matrix contains 150 dataset–model–fold tasks and 1,200 method rows. All primary comparisons use datasets as units after averaging five folds within each of TabICLv2 and TabPFN-3 and then averaging the two TFMs. CatBoost is descriptive only. Mean factorization TV is 0.06249 across the 20 primary cells, confirming measurable cross-query incompatibility.

Table 3 Dataset-blocked Selective-CoRe effects relative to arithmetic pooling across two TFMs and ten datasets. Negative values favor Selective CoRe.

Metric	Mean difference	95% bootstrap CI	Wins	p
Joint NLL	+0.001831	[+0.000548, +0.003301]	2/10	0.02734
Joint Brier	+0.000646	[+0.000288, +0.001059]	1/10	0.00586
ECE (15 bins)	+0.000096	[−0.002706, +0.003017]	6/10	1.00000
Marginal distortion	−0.003855	[−0.005563, −0.002212]	9/10	0.00391

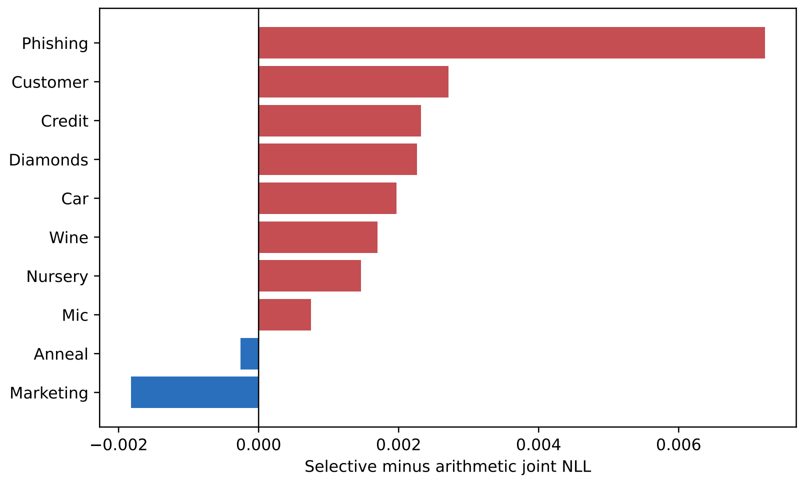

Fig. 3 Dataset-blocked Selective-CoRe joint-NLL effect relative to arithmetic pooling.

Selective CoRe therefore improves agreement with direct marginals but worsens both proper scores. Its NLL effect is positive in eight of ten datasets and under every leave-one-dataset-out mean. Hard CoRe nearly eliminates marginal distortion yet is worse than arithmetic by 0.02056 NLL on every primary dataset ($p = 0.00195$). Geometric pooling is descriptively 0.000492 better than arithmetic but not significantly so ($p = 0.7695$).

The effect reverses by upstream model: Selective CoRe is worse than arithmetic by 0.006694 NLL on TabICLv2 but better by 0.003031 on TabPFN-3. Their paired difference is 0.009725 ($p = 0.00977$; exploratory). The selection-family ablation also exposes finite-validation cost: pool-only mean NLL is 1.375591 across all active cells, compared with 1.376046 for the full family. Real-data validation-fraction results are non-monotone, so the controlled validation-size trend must not be generalized to this bounded setting.

8 Discussion

8.1 Coherence is a constraint, not an accuracy metric

The experiments support a distinction that is easy to lose in a consistency paper. Compatibility asks whether predictions can be represented by a single joint. Calibration asks whether stated probabilities match empirical frequencies. Proper scoring asks whether the full forecast assigns probability effectively to realized outcomes. Distributional distance to an exact posterior asks yet another question. These criteria can disagree. The independent baseline is coherent and can be poor; Adaptive Soft can improve log loss while not minimizing TV to truth; and a hard projection can preserve two accurate marginals or two inaccurate ones with equal mathematical success.

8.2 Selection is useful only with enough evidence

Unconditional projection is poorly matched to heterogeneous view reliability, but the real benchmark shows that a large selector is not automatically the solution. Searching 48 candidates with 52 validation labels can lose to arithmetic pooling. Raw directions and pools must remain available, while selector complexity must be matched to the evidence budget and upstream model.

8.3 Relation to label shift and OOD robustness

CoRe-TFM addresses incompatibility among simultaneous probability views; it is not designed to correct class-prior shift. DistPFN (Lee et al. 2026) is complementary rather than a baseline that CoRe subsumes. Similarly, OOD degradation (Loza et al. 2026) can change all four views together while leaving them relatively compatible. Future work should test whether distribution shift changes not only predictive performance but also the reliability ordering among direct and conditional views.

9 Limitations and threats to validity

First, the formulation is pairwise and categorical. Second, the real benchmark uses two TFMs; CatBoost is not a third foundation model. Third, every primary fold uses one deterministic constrained sample with at most 256 training observations, so independent sampling seeds and larger context sizes are required. Fourth, Marketing and Nursery contain extremely rare target classes and require an exclusion sensitivity. Fifth, the 48-candidate selector is large relative to 52 validation labels. Sixth, ECE is descriptive. Finally, the controlled mixture is a designed reliability stress test, not an estimate of naturally occurring TFM error regimes.

10 Reproducibility and data protocol

The repository at https://github.com/bnssaanirudh/core-tfm contains the exact Colab notebook, 1,200 fold-level rows, controlled replications, ablations, validation sensitivity, fold manifests, source and data hashes, package versions, hardware metadata, plots, and a checksum manifest. A repository-native audit command validates the matrix and regenerates dataset-blocked effects. The run used Python 3.13.15, PyTorch 2.11.0 with CUDA 12.8, TabICL 2.1.1, TabPFN 8.4.0, CatBoost 1.2.10, and a Tesla T4. Access credentials are never stored.

11 Conclusion

CoRe-TFM treats probability compatibility as a four-view reliability problem. Controlled experiments establish that marginal preservation helps only when the marginals deserve trust and that adaptive selection can work with adequate validation evidence. The bounded real-model benchmark supplies the counterexample needed for a credible conclusion: Selective CoRe reduces marginal distortion but is significantly worse than arithmetic pooling on NLL and Brier score, and its effect reverses between TabICLv2 and TabPFN-3. Reconciliation should therefore be deployed conditionally, with complexity matched to the reliability structure and validation budget. Sampling-seed, context-size, and rare-class robustness remain submission requirements.

Acknowledgements. [To be completed if applicable.]

Declarations

Funding. [Confirm funding statement before submission.]

Competing interests. [Author to confirm before submission.]

Ethics approval and consent to participate. Not applicable; the study uses synthetic data and public benchmark datasets.

Data availability. Synthetic data are generated from fully specified code and random seeds. Real benchmark datasets are publicly available through the source repositories/OpenML identifiers listed in Table 1. Final dataset snapshots and hashes will be archived with the experimental release.

Code availability. Research code and reproducibility scripts are available at https://github.com/bnssaanirudh/core-tfm. A versioned archival release will be created for the submitted manuscript.

Author contributions. Badampudi Agasthya Anirudh: conceptualization, methodology, software, experiments, analysis, visualization, writing, and project administration. [Revise if additional authors contribute before submission.]

Generative AI statement. [Complete in accordance with Springer Nature's current disclosure policy before submission. The author remains responsible for verification of all text, code, experiments, calculations, and references.]

References

Benamou JD, Carlier G, Cuturi M, et al (2015) Iterative bregman projections for regularized transportation problems. SIAM Journal on Scientific Computing 37(2):A1111–A1138. https://doi.org/10.1137/141000439

Chizat L, Peyre G, Schmitzer B, et al (2018) Scaling algorithms for unbalanced transport problems. Mathematics of Computation 87(314):2563–2609. https://doi.org/10.1090/mcom/3303

Costa JLDM, Popineau F, Rimmel A, et al (2026) High performance, low reliability: Uncertainty benchmarking for tabular foundation models. arXiv preprint arXiv:260528554 https://doi.org/10.48550/arXiv.2605.28554

Cuturi M (2013) Sinkhorn distances: Lightspeed computation of optimal transport. In: Advances in Neural Information Processing Systems

Gneiting T, Raftery AE (2007) Strictly proper scoring rules, prediction, and estimation. Journal of the American Statistical Association 102(477):359–378. https://doi.org/10.1198/016214506000001437

Google Research (2026) Introducing tabfm: A zero-shot foundation model for tabular data. Technical release, accessed August 2026

Grinsztajn L, Floge K, Key O, et al (2026) Tabpfn-3: Technical report. arXiv preprint arXiv:260513986 https://doi.org/10.48550/arXiv.2605.13986

Guo C, Pleiss G, Sun Y, et al (2017) On calibration of modern neural networks. In: Proceedings of the 34th International Conference on Machine Learning, pp 1321–1330

Kloetergens C, Yalavarthi VK, Schmidt-Thieme L, et al (2026) Do tabular foundation models agree with themselves? arXiv preprint arXiv:260806004 https://doi.org/10.48550/arXiv.2608.06004

Landsgesell J, Knoll P (2026a) Distributional regression with tabular foundation models: Evaluating probabilistic predictions via proper scoring rules. arXiv preprint arXiv:260308206 https://doi.org/10.48550/arXiv.2603.08206

Landsgesell J, Knoll P (2026b) Scoringbench: A benchmark for evaluating tabular foundation models with proper scoring rules. arXiv preprint arXiv:260329928 https://doi.org/10.48550/arXiv.2603.29928

Lee S, Lee J, Seo J, et al (2026) Mitigating label shift in tabular in-context learning via test-time posterior adjustment. arXiv preprint arXiv:260504363 https://doi.org/10.48550/arXiv.2605.04363

Loza M, Chushig-Muzo D, Milara E, et al (2026) Empirical evaluation of out-of-distribution performance of tabular foundation models. arXiv preprint arXiv:260726000 https://doi.org/10.48550/arXiv.2607.26000

Mure J (2017) Optimal compromise between incompatible conditional probability distributions, with application to objective bayesian kriging. arXiv preprint arXiv:170307233 https://doi.org/10.48550/arXiv.1703.07233

Qu J, Holzmuller D, Varoquaux G, et al (2026) Tabiclv2: A better, faster, scalable, and open tabular foundation model. arXiv preprint arXiv:260211139 https://doi.org/10.48550/arXiv.2602.11139

Young R (2026) On the conditioning consistency gap in conditional neural processes. arXiv preprint arXiv:260419312 https://doi.org/10.48550/arXiv.2604.19312